Cyclic marked trace with a Frobenius normalizer

Public G,Q -> cyclic graph trace

No nonrational torsion basis; r graph pairs

Two normalizations -> same marked point action

Cyclic kernel denominator, or compact (X^p-X)^2 with factor4*y(U)^2

Balanced rational merge -> coefficient construction

Five polynomial products per node; O(M(r)log r) after graph/weights

P-192 Frobenius normalizer -> compressed map

Actual quadratic-field map checked; paired weighted trace remains next

Construction, 2026-11-08. The marked principal coefficient from Round55 can be obtained from the cyclic rational graph Gamma=<{(G,Q)}> alone. The construction takes the public curve, G,Q and a quadratic shift; it needs no point in the nonrational r-torsion eigenspace. Its dense numerator has degree at most r instead of r^2. A second normalization uses the compact polynomial X^p-X and replaces the division-polynomial derivative by4*y(U)^2. P-192 remains the primary target. [pcc:unverified]

Domain. Let E/F_p:y^2=x^3+ax+b, p>3, with G of odd prime order r different from p, and Q=[x]G nonzero. The constructor receives Q, never x. Choose R0 in E(F_(p^2)) with2R0 nonrational. Define C_Gamma(P,R0)=sum_(j=0..r-1) x(P+jG)*x(R0+jQ). This trace is invariant under Gamma. Its global quotient is by Gamma; the bridge below concerns the local marked coefficient. When Gamma is the rational part of the full mixed kernel H, its principal coefficient at P=-U agrees with that of the full trace: each has exactly the term indexed by U as a pole there. The other terms of H are regular at that point. [pcc:unverified]

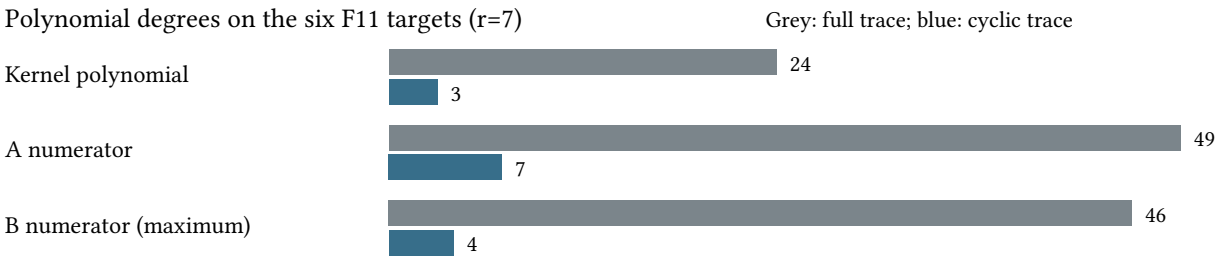
Degree reduction. Let f_G(X)=product_(j modulo sign, j!=0)(X-x(jG)), monic of degree(r-1)/2, D_G=f_G^2, and N_G=D_G*C_Gamma. Pair j with -j and use the addition formula from Round55 to obtain N_G=A(X)+Y*B(X), deg A<=r, deg B<=r-3. At U=(u,v)!=O in <G>, N_G(-U)=4v^2*f_G'(u)^2*x(R0+[x]U). This is a written polynomial argument; the native controls reconstruct every coefficient from the public pairs (jG,jQ). [pcc:unverified]

Derivative-free normalization. Put H_p(X)=X^p-X, D_0=H_p^2 and N_0=D_0*C_Gamma. Every kernel x-coordinate lies in F_p, so D_G divides D_0 and N_0 is a polynomial A_0+Y*B_0, with deg A_0<=2p+1, deg B_0<=2p-2. At X=u, H_p(X)=(X-u)^p-(X-u), hence (H_p/(X-u))(u)=-1. The unique pole term gives N_0(-U)=4v^2*x(R0+[x]U). Equivalently its individual terms use the compact factor ((X-u)^(p-1)-1)^2; no expanded p-degree polynomial is needed for that factor. The current complete sum still enumerates the cyclic kernel. [pcc:unverified]

Point extraction and normalization map. Divide by4v^2, lift the resulting x-coordinate over F_(p^2), and subtract R0. The two candidates are [x]U and -[x]U-2R0, exactly one rational. This gives the same marked action as Round55. There is also an explicit compressed normalizer: phi=1-pi, where pi is Frobenius. Its x-coordinate is ((Y^p+Y)/(X^p-X))^2-X^p-X. At P=-U, its leading coefficient in X-u is4v^2, so the local quotient C_Gamma/x(phi(P)) has limit x(R0+[x]U). This last statement is a local quotient, not permission to substitute into an uncancelled0/0 expression. [pcc:unverified]

Full-order specialization and P-192. When <G>=E(F_p), phi has degree r and this kernel. Its differential is1, so its normalized Velu quotient has codomain E itself. Writing F=X^3+aX+b, the raw x-map numerator is X^(2p+1)+X^(p+2)+aX^p+aX+2b+2F^((p+1)/2), with denominator(X^p-X)^2. These powers give a compact arithmetic expression. At rational U, its numerator is4v^2 and its denominator begins with(X-u)^2. Native full-width checks at the saved P-192 points G,Q,2G reproduce those constants. At G+S,Q+S,2G+S for the saved quadratic-twist point S, the explicit Frobenius-difference map equals[2]S and matches the coordinate formula. This constructs and evaluates the normalizer; the P-192 cyclic trace is not evaluated. [pcc:unverified]

Exact controls. The frozen panel uses all six targets and six inputs on E/F11:y^2=x^3+2x+7, r7, and all eighteen targets and eighteen inputs on E/F17:y^2=x^3+2x+2, r19. Both use the declared quadratic shift6S with S of order17. The producer reconstructs the cyclic and raw numerators, proves their polynomial factor identity by exact coefficients, checks the full Velu/Frobenius rational-map identity by polynomial cross multiplication, evaluates72 direct traces and360 marked actions, and rejects360 altered outputs. The separate native checker enumerates the complete quadratic-field curves and uses line intersections for addition, a packed field representation, and an unpaired reconstruction of all744 numerator coefficients. It checks all360 actions and72 traces. A further36 checks equate the cyclic principal values with the earlier full-trace records. All panel outcomes, including the two degree3 rather than4 B-polynomials over F11, remain in certificate.json. [pcc:unverified]



Unit: polynomial degree; all six targets. Native records, rounds55-56. No timing inference.

Cost model. Successive addition builds the public cyclic graph in O(r) point additions, as implemented by the reference checker; the producer instead uses O(r*log r) additions through individual binary multiples. A simple dense f_G and numerator builder uses O(r^2) field operations and O(r) coefficient storage; a Horner action uses O(r) field operations. The earlier dense full-torsion builder used O(r^4) and O(r^2) storage. These are derived costs of the stated algorithms, not timing measurements. The raw version uses O(r*p) simple coefficient work and O(p) storage, with an explicit4v^2 normalization. Reusable G-only data and every Q-dependent enumeration/build step must be charged separately. For d auxiliary powers, retain T_setup+(d-1)*T_action+T_Cheon+T_generator_setup. The fast normalizer does not supply a fast weighted trace. [pcc:unverified]

Balanced builder. For each pair +/-U, form a leaf with denominator d_U=(X-x(U))^2 and the degree2 A and constant B numerators from the addition formula. Merge (d1,a1,b1) and (d2,a2,b2) into (d1*d2,a1*d2+a2*d1,b1*d2+b2*d1), then add the identity term x(R0)*X*D_G at the root. This uses five polynomial products per internal node. A balanced tree takes O(M(r)*log r) field operations under the standard polynomial multiplication regularity assumption, after graph and weight construction; its output has O(r) coefficients. The native checker uses schoolbook M(n)=O(n^2), reconstructs all24 numerator pairs by this second algorithm, and compares every coefficient with both the unpaired reconstruction and the saved producer record. No fast native backend or timing claim is supplied. [pcc:unverified]

CM-invariant link and next construction. phi=1-pi is an explicit principal CM relation of norm r. It annihilates the rational subgroup, and is used here as a pole normalizer. The construction applies to each supplied prime-order curve; no small class number or L-value is assumed. To exploit those invariants, investigate a cheaper representation of the paired weights x(R0+jQ), using the costed horizontal relations rather than only the unmarked curve identity. compression-audit.json specifies the next weighted-resultant interface. It retains the sign coupling between jG and jQ and charges joint-coordinate interpolation; a one-coordinate kernel evaluator alone does not implement it. [pcc:unverified]

Formal scope. lean/Fiber.lean proves the abstract unique-rational-fiber step from explicit two-point-fiber and disjointness hypotheses, using Lean4.19.0 core and no mathematical library. The elliptic-curve binding, nonrationality argument, trace identity and local coefficient proof remain written arguments; proof-map.json separates them. This is partial formalization. Native checks and the Lean lemma are producer evidence; no trusted signed PCC verdict is asserted. [pcc:unverified]

Sources checked. BFLS, Faster computation of isogenies of large prime degree, arXiv2003.10118, Lemma4.3, Lemma4.9, Algorithm2 and Theorem4.11: https://arxiv.org/pdf/2003.10118 . Its single-coordinate kernel-product result motivates the weighted-resultant interface; no complexity bound for the new paired trace is imported. MIT18.783 Lecture5 gives normalized Velu formulas: https://math.mit.edu/classes/18.783/2025/LectureSlides5.pdf . Egami1995, https://www.numdam.org/item/CM_1995__99_1_99_0.pdf , sections2-3 treats complex elliptic Dedekind reciprocity; it supplies no finite-field evaluator for this trace. Earlier rounds19,25,35 and55 retain the CM, ray-class, relation and principal-part evidence. Sources checked (KB): unavailable tools and absent migrated wrapper; no novelty or absence claim follows from this search. [pcc:unverified]